CIS-445 Machine Learning

Weka – Individual Project 2 Report

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The tutorial is worth 100 points.

Objectives: Learn about Weka, the Machine Learning Workbench

This project contains two parts: Part A and Part B. In both parts you will use Weka Explorer and Weka Experimenter. In Part A you will work with the `diabetes.arff` dataset and essentially follow the given instructions. In Part B, however, you will work with `heart.csv` dataset and, partially on your own, explore the performance of several models. Some parts of the project describe Weka and machine learning algorithms. You need to read these parts. There are also other parts which you have to work hands-on, step by step, in Weka. Note that you need to copy some of the results and answer the questions which I ask in the project and paste them into this document so that I can check how well you followed all steps in the project. Because this is the second version of the Weka – Individual Project 2, I would appreciate <u>your comments</u> and suggestions regarding the organization, clarity, spelling, content, and/or anything else.

Please note that this file which now contains 2 pages may grow to a few more pages after you copy and paste here the screen images, tables, answers to questions, etc. from Weka – Project 2 file.

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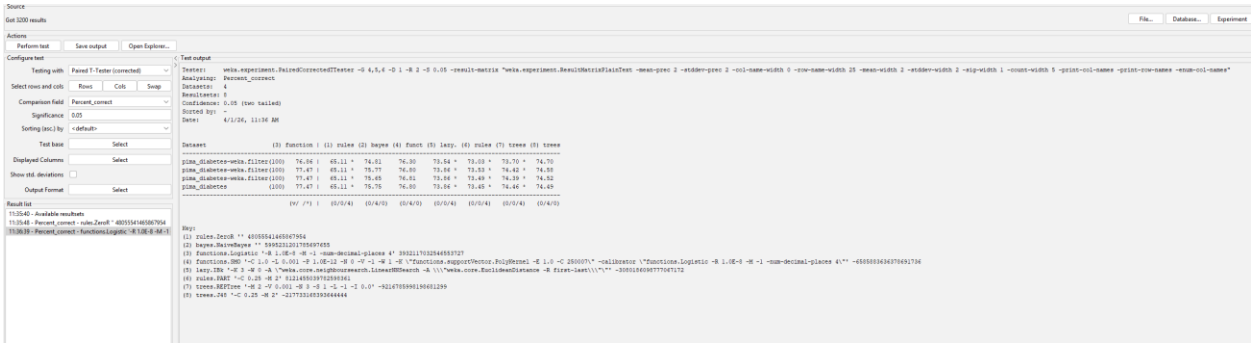
Copy and paste below the screen shot (Listing 2) from **<u>your</u>** run (**<u>not</u>** from my run above) in to this Weka – Individual Project 2 Report and discuss the overall classification accuracy rates. Compared to the logistic regression model, used as the baseline here, the performance of which models is statistically worse (*), better (v) or the same (/ /)?



None of them are better than Logistic, Only SMO is comparable to Logistic and all of the rest are worse.

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Copy and paste the screen shots for (a) through (d) below into this Weka – Individual Project 2 Report. The format of each screen shot should be similar to Listing 2 in Part A.

(a) Paste below the screen shot depicting the overall classification accuracy rates (*Percent correct*). Compare and briefly discuss *Percent correct* results in terms of the best/worst/same models for 6 models and 5 datasets.

```
Test output

Tester:     weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matri
Analysing:  Percent_correct
Datasets:   5
Resultsets: 6
Confidence: 0.05 (two tailed)
Sorted by:  -
Date:       4/1/26, 12:28 PM


Dataset                   (1) function | (2) funct (3) funct (4) meta. (5) lazy. (6) trees
-----------------------------------------------------------------------------------------
'heart-weka.filters.unsup (10)   85.73 |   85.30     86.17     85.08     84.20     85.08
'heart-weka.filters.unsup (10)   84.21 |   83.88     84.97     84.10     84.53     84.64
'heart-weka.filters.unsup (10)   86.28 |   86.06     85.52     84.86     86.38     86.28
'heart-weka.filters.unsup (10)   86.28 |   86.06     85.52     84.86     86.38     86.28
'heart-weka.filters.unsup (10)   86.28 |   86.06     85.52     84.75     86.38     86.28
-----------------------------------------------------------------------------------------
                                 (v/ /*) |   (0/5/0)   (0/5/0)   (0/5/0)   (0/5/0)   (0/5/0)


Key:
(1) functions.Logistic '-R 1.0E-8 -M -1 -num-decimal-places 4' 3932117032546553727
(2) functions.SMO '-C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K \"functions.supportVector.
(3) functions.MultilayerPerceptron '-L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H a -D' -59906078
(4) meta.Bagging '-P 100 -S 1 -num-slots 1 -I 10 -W trees.REPTree -- -M 2 -V 0.001 -N 3 -S
(5) lazy.IBk '-K 10 -W 0 -A \"weka.core.neighboursearch.LinearNNSearch -A \\\"weka.core.Euc
(6) trees.J48 '-C 0.25 -M 2' -217733168393644444
```

SMO is often the best, followed Bagging, IBk and Multilayer Perception, and then J48 is the lowest

(b) Paste below the screen shot depicting the classification accuracy rates for 1's - patients with heart failure (*True_positive_rate*). Compare and briefly discuss *True_positive_rate* results in terms of the best/worst/same models for 6 models and 5 datasets.

```
Test output
Tester:     weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matri
Analysing:  True_positive_rate
Datasets:   5
Resultsets: 6
Confidence: 0.05 (two tailed)
Sorted by:  -
Date:       4/1/26, 12:31 PM


Dataset                    (1) functio | (2) func (3) func (4) meta (5) lazy (6) tree
-----------------------------------------------------------------------------------------
'heart-weka.filters.unsup (10)   0.81 |   0.80     0.81     0.79     0.83     0.79
'heart-weka.filters.unsup (10)   0.81 |   0.78     0.80     0.79     0.82     0.77
'heart-weka.filters.unsup (10)   0.82 |   0.83     0.80     0.79     0.83     0.82
'heart-weka.filters.unsup (10)   0.82 |   0.83     0.80     0.79 *   0.83     0.82
'heart-weka.filters.unsup (10)   0.82 |   0.83     0.80     0.79     0.83     0.82
-----------------------------------------------------------------------------------------
                                (v/ /*) |  (0/5/0)  (0/5/0)  (0/4/1)  (0/5/0)  (0/5/0)


Key:
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(4) meta.Bagging '-P 100 -S 1 -num-slots 1 -I 10 -W trees.REPTree -- -M 2 -V 0.001 -N 3 -S
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(6) trees.J48 '-C 0.25 -M 2' -217733168393644444
```

IBk and SMo are the strongest, but SMO is less consistent. Bagging and J48 perform the worst and MLP is in the middle.

(c) Paste below the screen shot depicting the classification accuracy rates for 0's – healthy patients (*True_negative_rate*). Compare and briefly discuss *True_negative_rate* results in terms of the best/worst/same models for 6 models and 5 datasets.

```
Test output
Tester:     weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matri
Analysing:  True_negative_rate
Datasets:   5
Resultsets: 6
Confidence: 0.05 (two tailed)
Sorted by:  -
Date:       4/1/26, 12:33 PM


Dataset                    (1) functio | (2) func (3) func (4) meta (5) lazy (6) tree
-----------------------------------------------------------------------------------------
'heart-weka.filters.unsup (10)   0.90 |   0.90     0.90     0.90     0.85     0.90
'heart-weka.filters.unsup (10)   0.87 |   0.88     0.89 v   0.89     0.86     0.91 v
'heart-weka.filters.unsup (10)   0.89 |   0.89     0.90     0.90     0.89     0.90
'heart-weka.filters.unsup (10)   0.89 |   0.89     0.90     0.90     0.89     0.90
'heart-weka.filters.unsup (10)   0.89 |   0.89     0.90     0.89     0.89     0.90
-----------------------------------------------------------------------------------------
                                (v/ /*) |  (0/5/0)  (1/4/0)  (0/5/0)  (0/5/0)  (1/4/0)


Key:
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(6) trees.J48 '-C 0.25 -M 2' -217733168393644444
```

J48 performs the best here

SMO, MLP and Bagging are all the mid tier
And IBk is the weakest

(d) Paste below the screen shot depicting the areas under the ROC curves (*Area_under_ROC*). Compare and briefly discuss *Area_under_ROC* results in terms of the best/worst/same models for 6 models and 5 datasets. Recall that Receiver Operating Characteristics (ROC) curves depict the global performance of the models for cut-offs from within the range [0,1]. The standard cut-off for classification problems is 0.5.

```
Test output
Tester:     weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matri
Analysing:  Area_under_ROC
Datasets:   5
Resultsets: 6
Confidence: 0.05 (two tailed)
Sorted by:  -
Date:       4/1/26, 12:33 PM


Dataset                   (1) functio | (2) func (3) func (4) meta (5) lazy (6) tree
-----------------------------------------------------------------------------------
'heart-weka.filters.unsup (10)   0.92 |   0.85 *   0.92     0.91     0.91     0.86 *
'heart-weka.filters.unsup (10)   0.91 |   0.83 *   0.91     0.91     0.91     0.88
'heart-weka.filters.unsup (10)   0.93 |   0.86 *   0.93     0.92     0.92     0.88 *
'heart-weka.filters.unsup (10)   0.93 |   0.86 *   0.93     0.92     0.92     0.88 *
'heart-weka.filters.unsup (10)   0.93 |   0.86 *   0.93     0.92     0.92     0.88 *
-----------------------------------------------------------------------------------
                              (v/ /*) |  (0/0/5)  (0/5/0)  (0/5/0)  (0/5/0)  (0/1/4)


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(6) trees.J48 '-C 0.25 -M 2' -217733168393644444
```

MLP is the strongest
Followed by Bagging and IBk
J48 and SMO are the worst

Discuss the results in terms of the best/worst models by comparing Percent correct, True_negative_rate, True_postive_rate, and Area_under_ROC combined. Often you have to consider several measures combined, not just one measure, to choose the best model.

I would choose MLP, because when you average out all of their results from the 3 datasets, it consistently scores the highest, and is number the lowest.
The worst model was J48, which was consitently rated the worst.

Does normalization, standardization, and two attribute reduction techniques, which you applied, improve the performance of the models? Look at the classification accuracy rates for these particular models and datasets.

Yes, most data performed better after preprocessing the data. It was most seen in MLP, and the percent correct for all models got better as the data became more processed.

Describe below what you did in Part B of the project.

I created each of the 5 files, Nominal, Normalize, Standardize, CFS Subset and Gain Ratio,
Added the 6 Classifaction Algorithms, adjusted them as needed.
Took Screenshots of each of the Expierements I ran
Answered each of the questions.

I would appreciate your comments and suggestions regarding the organization, clarity, spelling, content, and/or anything else. Provide the comments below. (optional)